Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Agentic AI Applications with Claude Code

TGS Ref No: TGS-2025052468

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 2.2

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	27 Jun 2026	First version — full-day (9:00 AM–6:00 PM) lesson plan mapped to the three topics and activities	Dr. Alfred Ang
1.1	27 Jun 2026	One-day schedule with 1-hour assessment at 5:00 PM (30 min WA + 30 min PP), shortened breaks, and courseware/assessment on the LMS	Tertiary Infotech Academy Pte Ltd
1.2	27 Jun 2026	Aligned Topic 1 to the 7-step build workflow (Activity 1a/1b) and added the Hooks activity	Tertiary Infotech Academy Pte Ltd
1.3	8 Jul 2026	Added a “Setup Claude Code in VS Code” slide (now slide 19) and realigned all slide-number references to the updated 65-slide deck (v13.0)	Tertiary Infotech Academy Pte Ltd
1.4	2 Sep 2026	Added Claude for Chrome & Claude for Science to the Claude Product Family slide; added a “Plan” step to the Agentic Loop in Action slide; added a new “SKILL.md vs CLAUDE.md” slide. Slide references updated.	Tertiary Infotech Academy Pte Ltd
1.5	2 Sep 2026	Added a Practice Exam Portal slide (exams.tertiaryinfotech.com) and an LMS screenshot on the Courseware & Assessment slide; reworked Activity 6 to the floating WhatsApp widget with a 10-second auto-open hook, suggested queries and voice activation.	Tertiary Infotech Academy Pte Ltd
1.6	4 Sep 2026	Added a “User Level	Tertiary Infotech

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		vs Project Level" slide (slide 42) comparing where commands, MCP tools, skills, CLAUDE.md and hooks are stored at user level (applies to all projects) versus project level (one project only); realigned all slide-number references to the 68-slide deck (v14.0)	Academy Pte Ltd
1.7	9 Sep 2026	Added a "What Triggers What" key-concepts summary slide (new slide 64) covering CLAUDE.md, skills, tools, commands, hooks and sub agents and their triggers; expanded the SKILL.md vs CLAUDE.md comparison.	Tertiary Infotech Academy Pte Ltd
1.8	9 Sep 2026	Corrected Topic 1.2 / Activity 1 slide references (31–38 and 25–30, 39). Added the Lab Materials section covering the four restructured lab folders (Lab 1, Lab 1b, Lab 2, Lab 3).	Tertiary Infotech Academy Pte Ltd
1.9	9 Sep 2026	Recomputed every slide reference against deck v15 after the Lab Materials slide insert shifted numbering by one: Topic 1.2 = 32–39, Activity 1 = 31 and 40, and all other schedule rows and topic headings realigned.	Tertiary Infotech Academy Pte Ltd
2.0	29 Oct 2026	Recomputed every slide reference against deck v21 (77 slides) after seven setup/GitHub	Tertiary Infotech Academy Pte Ltd

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		screenshot slides were added and the agentic-loop slides moved to follow Harness Engineering (slides 19–22) and the setup walkthrough moved into Activity 1 after the 7 Steps (29–31): Topic 1 = 14–49 (Activity 1 = 28–32, 37, 46–49), Topic 2 = 50–59, Topic 3 = 60–73, closing = 72–77. Replaced the retired Mini Capstone with Activities 6–7 to match the deck.	
2.1	29 Oct 2026	Recomputed slide references against deck v22 (79 slides) after two hook-example slides were added after Automating with Hooks (slides 70–71): Topic 3 = 60–75, Topic 3.2/3.3 = 64–71, Activities 6–7 = 72–73, closing = 74–79.	Tertiary Infotech Academy Pte Ltd
2.2	31 Oct 2026	Slide references recomputed for deck v23 (83 slides; new slides 19–20, 64, 68): Topic 1 = 14–51, Topic 2 = 52–61, Topic 3 = 62–79, closing = 78–83. Added the seven parts of an AI agent and skill vs sub agent front matter.	Tertiary Infotech Academy Pte Ltd

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Learning Outcomes

By the end of this course, learners will be able to:

1. Explain the agentic loop and apply context-engineering techniques in Claude Code.
2. Create custom slash commands and connect MCP tools to extend Claude Code.
3. Use agent skills, sub agents and hooks to automate specialised tasks.
4. Build, deploy, test, secure and optimise a real website end-to-end with Claude Code.

Daily Schedule (9:00 AM – 6:00 PM, one-day class)

This is a **one-day class**, so the **1-hour assessment runs from 5:00 – 6:00 PM** and all teaching content finishes by 5:00 PM. Breaks are shortened (10-minute breaks, 45-minute lunch) to make room.

Time	Duration	Topic / Activity	Method	Slides
9:00 – 9:30	30 min	**Welcome & Administration** — Digital attendance (AM), trainer introduction, learner introductions, ground rules, course outline	Lecture / Discussion	1–13
9:30 – 10:30	60 min	**Topic 1.1 Claude Code fundamentals — What makes an AI agent (model + harness + tokens; seven parts); how it works (agentic loop + built-in tools); Setup (Terminal, VS Code, Desktop) & pricing; permission modes & bypass (VS Code and Desktop settings)**	Lecture / Demo	14–29, 35–38
10:30 – 10:40	10 min	**Morning Break**	—	—
10:40 – 11:20	40 min	**Topic 1.2 Context, memory & sessions** — context window & /context, CLAUDE.md vs auto memory, managing sessions, scheduled tasks, /goal	Lecture / Demo	40–47
11:20 – 12:30	70 min	**Activity 1 — Build & Deploy a Website — Build a responsive software-dev site (hero, testimonials, enquiry form) after setting up the VS Code extension / Desktop app, create a GitHub repository and push with Claude, deploy on GitHub Pages via Actions, add README**	Hands-on	30–34, 39, 48–51
12:30 – 1:15	45 min	**Lunch Break**	—	—

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1:15 – 1:45	30 min	**Topic 2.1 Custom Commands** — Slash commands, .claude/commands, arguments; **Activity 2 — Create /gitpush command**	Lecture / Hands-on	52–56
1:45 – 2:45	60 min	**Topic 2.2 MCP Tools** — MCP architecture, connecting servers & scopes, Playwright MCP; **Activity 3 — screenshot site & push; Activity 3B — test enquiry form to email**	Lecture / Hands-on	57–61
2:45 – 2:55	10 min	**Afternoon Break**	—	—
2:55 – 3:40	45 min	**Topic 3.1 Agent Skills** — SKILL.md and its front matter (name, description), project skills, installing skills; **Activity 4 — Install frontend-design & seo-audit skills, optimise site**	Lecture / Hands-on	62–66
3:40 – 4:25	45 min	**Topic 3.2 Sub Agents & 3.3 Hooks — Creating sub agents and their front matter (name, description, tools, skills, model), hook events, hooks in action (10-second pop-up, voice greeting on chat click); Activity 5 — Security sub agent + lead-magnet UI/UX sub agent**	Lecture / Hands-on	67–75
4:25 – 4:50	25 min	**Activities 6–7 — Hooks (floating WhatsApp widget), final testing, optimisation & deployment**	Hands-on	76–77
4:50 – 5:00	10 min	**Summary, Q&A & Closing — Key-concepts recap (what triggers what), recap, digital attendance, TRAQOM survey & certification**	Discussion	78–83
5:00 – 5:30	30 min	**Written Assessment (WA)** — open book, on the LMS	Assessment	—
5:30 – 6:00	30 min	**Practical Performance (PP)** — open book, on the LMS	Assessment	—

The 1-hour assessment (30 min WA + 30 min PP) runs from 5:00 PM and the class ends at 6:00 PM. Breaks are shortened (10-minute breaks, 45-minute lunch) to fit the assessment. Courseware and the assessment are available on the LMS at https://lms-tms.tertiaryinfotech.com/. Timings may be adjusted slightly to suit class pace.

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Topic-by-Topic Breakdown

Topic 1 — Claude Code Fundamentals (≈ 1 hr 15 min + Activity 1) · Slides 14–51

- What makes an AI agent: model + harness + tokens; the seven parts (model, tools, skills, memory, knowledge base, instructions, runtime controls) and where each lives in Claude Code
- How Claude Code works: the agentic loop and built-in tools
- Setup: Terminal, IDE (VS Code) and the Desktop app; pricing
- Context engineering & the context window (/context, /compact, /clear)
- Memory files: CLAUDE.md vs auto memory (/init, /memory)
- Permission modes and the dangers of bypass permissions; allowing bypass mode in VS Code and Claude Desktop
- Managing sessions, scheduled tasks (Routines), and /goal
- **Activity 1: Build and deploy a website — set up the VS Code extension (secondary side bar) or the Desktop app, then create a GitHub repository and push with Claude (Step 5)**

Topic 2 — Tools and Commands (≈ 1 hr 30 min incl. activities) · Slides 52–61

- Custom slash commands and arguments
- **Activity 2:** Create a /gitpush custom command
- MCP tools and architecture; Playwright MCP
- **Activity 3:** Install Playwright MCP, screenshot site, push
- **Activity 3B:** Test the enquiry form with Playwright; confirm enquiries reach your email

Topic 3 — Skills, Agents & Hooks (≈ 1 hr 30 min incl. Activities 6–7) · Slides 62–79

- Agent skills and installation; inside a SKILL.md — the name and description front matter decide when a skill fires
- **Activity 4:** Install & run frontend-design and seo-audit skills
- Sub agents — front matter: name, description, tools, skills and model — and hooks; hooks in action — a 10-second pop-up and a voice greeting on chat click
- **Activity 5:** Security and UI/UX (lead-magnet) sub agents
- **Activity 6: Floating WhatsApp widget with a 10-second auto-open hook, suggested queries and voice activation; Activity 7: test, optimise and deploy**
- **Key concepts summary: what triggers what — CLAUDE.md (every session), skills and tools (on demand), slash commands (manual), hooks (by event) and sub agents (by delegation); SKILL.md vs CLAUDE.md**

Lab Materials

- Lab 1 (Topic 1) — labs/lab1-seven-steps/: the 7-step build workflow, one file per step, plus a reference site and three sample sites.
- Lab 1b (Topic 2) — labs/lab1b-commands-mcp/: build a /publish custom command, install Playwright MCP, screenshot the live site into the README.
- Lab 2 (Topic 3) — labs/lab2-skills/: install the cybersecurity-analyst, agent-kanban and ui-ux-pro-max skills and revamp the Lab 1 site.

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- Lab 3 (Topic 3) — labs/lab3-hooks/: a completion hook, the WhatsApp widget, an add-task sub agent, a security-scan sub agent writing JSON, and /loop.
- Labs repository: https://github.com/tertiarycourses/TGS-2025052468-Claude-Code

Resources Required

- Computers with internet access (per learner)
- Claude Code installed and signed in
- GitHub accounts; Node.js installed
- Projector / shared screen for demonstrations
- Course slides and this learner guide

Assessment

- **Written Assessment (WA):** 30 min — open book, on the LMS
- **Practical Performance (PP):** 30 min — open book, on the LMS
- Assessment starts at 5:00 PM and the class ends at 6:00 PM.
- **Courseware & assessment:** available on the LMS at https://lms-tms.tertiaryinfotech.com/
- **Funding criteria:** minimum 75% attendance and assessed as *Competent*

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